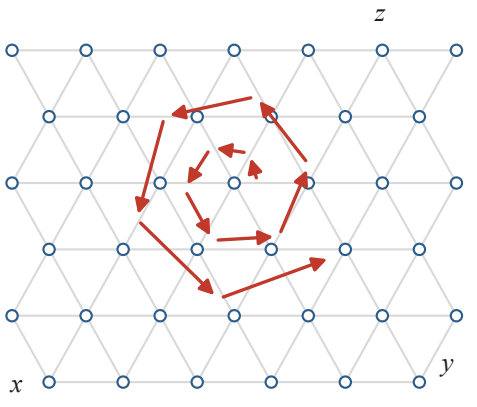
The Gauge Lift

1 6D Lattice Deformation



Phason displacement $u_\perp$ in internal space $E_\perp$.

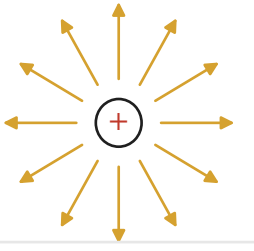
2 Lift Operation: Curl Extraction

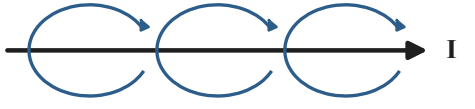
$$F_{\mu\nu} \propto \nabla \times u_\perp$$

Internal space ($E_\perp$) → external gauge field.

Curl extraction selects the transverse vector content.

3 Electromagnetic Field $F_{\mu\nu}$ in 3D Spacetime





B (magnetic field)

$$B = \nabla \times u_\perp$$

$$E = \partial_t u_\perp$$

Spin-1: vector field structure inherited from curl operation. Massless: $U(1)$ gauge protection.